What One and Two L/R/U/D Polarimeters Can Determine from a Spin–1 Beam

Concise DPOLAR analysis note

Abstract

A general spin–1 density matrix has eight real polarization degrees of freedom: three vector components and five tensor components. A single ideal L/R/U/D polarimeter does not determine all eight. With several polar scattering angles, controlled normalization, and non-degenerate analyzing powers, one station can determine at most six combinations. Its exact cardinal-geometry nulls are the longitudinal vector polarization p_z and the tensor component p_{xy}. A second station can determine all eight only if known spin transport rotates both null directions into combinations observed at the other station. If particles follow different paths, the transport is an ensemble of rotations rather than a single rotation; this can reduce polarization, mix components, and introduce additional nuisance parameters that lower identifiability.

1 The short answer

The spin–1 density matrix can be written in terms of

$$\boldsymbol{p} = (p_x, p_y, p_z), \qquad \boldsymbol{P} = \begin{pmatrix} p_{xx} & p_{xy} & p_{xz} \\ p_{xy} & p_{yy} & p_{yz} \\ p_{xz} & p_{yz} & p_{zz} \end{pmatrix}, \qquad p_{xx} + p_{yy} + p_{zz} = 0. \tag{1}$$

Thus there are $3 + 5 = 8$ real polarization degrees of freedom.

Table 1: Maximum structural information under ideal calibration. "Multiple θ" means at least two polar-angle groups whose analyzing powers have different angular dependence.

Configuration	Maximum rank	Determinable	Remaining exact nulls
One station, one θ, known normalization	4/8	one common mode, two first harmonics, one diagonal harmonic	p_z, p_{xy}, and two vector–tensor first-harmonic mixtures
One station, multiple θ, shared normalization	6/8	$p_x, p_y, p_{xz}, p_{yz}, p_{xx} - p_{yy}, p_{zz}$	p_z and p_{xy}
Two identical stations, aligned transport	6/8	the same six combinations with better precision	the same two null directions
Two stations, generic non-longitudinal relative rotation	up to 8/8	all vector and tensor components	none, if the transported null spaces do not overlap
Two stations, relative rotation only about z	up to 7/8	the tensor null may be lifted	p_z remains invariant and invisible

These are upper bounds. Unknown luminosities, independent detector efficiencies, backgrounds, or uncertain spin transport can reduce the effective rank.

2 Why one station has only six independent combinations

In the local beam frame, the repository cross-section convention is

$$
\begin{aligned}
\frac{\sigma(\theta,\phi)}{\sigma_0(\theta)} &= 1 + \frac{3}{2}(p_x \sin\phi + p_y \cos\phi) A_y(\theta) \\
&\quad + \frac{2}{3}(p_{xz}\cos\phi - p_{yz}\sin\phi) A_{xz}(\theta) \\
&\quad + \frac{1}{6}\left[(p_{xx}-p_{yy})\cos 2\phi - 2p_{xy}\sin 2\phi\right]\left[A_{xx}(\theta) - A_{yy}(\theta)\right] \\
&\quad + \frac{1}{2} p_{zz} A_{zz}(\theta).
\end{aligned} \tag{2}
$$

The four azimuthal harmonics have the interpretation

Harmonic	Polarization combination	L/R/U/D information
common	$p_{zz}A_{zz}$	absolute or cross-angle rate
$\cos\phi$	$\frac{3}{2}p_y A_y + \frac{2}{3}p_{xz}A_{xz}$	$L - R$
$\sin\phi$	$\frac{3}{2}p_x A_y - \frac{2}{3}p_{yz}A_{xz}$	$U - D$
$\cos 2\phi$	$(p_{xx}-p_{yy})(A_{xx}-A_{yy})/6$	$(L+R)-(U+D)$
$\sin 2\phi$	$-p_{xy}(A_{xx}-A_{yy})/3$	zero at cardinal azimuths

At one polar angle, the first harmonics contain inseparable vector–tensor pairs: (p_y, p_{xz}) and (p_x, p_{yz}). Several polar angles separate each pair if the angular responses are non-collinear, schematically

$$
\det\begin{pmatrix} A_y(\theta_1) & A_{xz}(\theta_1) \\ A_y(\theta_2) & A_{xz}(\theta_2) \end{pmatrix} \neq 0. \tag{3}
$$

Similarly, p_{zz} can be separated from one shared luminosity scale only if the acceptance-averaged A_{zz} responses differ between angular groups. Independent normalizations for every angle remove this information.

Two directions remain invisible even with many polar angles:

1. p_z does not occur in Eq. (2); the reaction is sensitive only to transverse vector polarization.
2. p_{xy} multiplies $\sin 2\phi$, which is exactly zero for $\phi = 0°, 90°, 180°, 270°$ and for symmetric finite openings centered on those azimuths.

For the repository's current tensor-only calculation, vector polarization is excluded from the fit. The planned two-angle proton-singles response then has rank 4/5: it determines $p_{xx} - p_{yy}$, p_{xz}, p_{yz}, and p_{zz}, while p_{xy} is the exact null direction. This is consistent with the general 6/8 result after the two transverse vector components are restored.

3 What a second station adds

Let the upstream density-matrix parameters be a common reference vector $\boldsymbol{x}_0 = (\boldsymbol{p}_0, \boldsymbol{q}_0)$, with three vector and five tensor coordinates. Known spin transport to station s acts as

$$
\boldsymbol{p}_s = R_s \boldsymbol{p}_0, \qquad \boldsymbol{P}_s = R_s \boldsymbol{P}_0 R_s^{\mathsf{T}}, \qquad \boldsymbol{q}_s = T^{(2)}(R_s)\boldsymbol{q}_0. \tag{4}
$$

If H_s is the local station response, the combined response is

$$
J_{\text{combined}} = \begin{pmatrix} H_1 M_1 \\ H_2 M_2 \end{pmatrix}, \qquad M_s = R_s \oplus T^{(2)}(R_s), \tag{5}
$$

and its null space is

$$
\boxed{\mathcal{N}_{\text{combined}} = \ker(H_1 M_1) \cap \ker(H_2 M_2)}. \tag{6}
$$

Equation (6) is the practical criterion.

- If the two stations have the same geometry and aligned spin transport, they measure the same six-dimensional subspace. Counts increase, but rank does not.
- A relative rotation about z can rotate the tensor p_{xy} null into a visible diagonal tensor component. However, it leaves p_z unchanged, so the full density matrix still has at least one null direction.
- A generic rotation containing an x or y component can rotate p_z into p_x or p_y and can simultaneously rotate p_{xy} into visible tensor components. Full rank eight is possible if both transported nulls are complementary and the nuisance model does not absorb them.

For the immediate nominal pure-p_{yy} experiment, the parameter space is much smaller than eight dimensions. One after-SRC station is sufficient for tensor magnitude and stability monitoring and for the first-order tilt that generates p_{yz}. The orthogonal tilt generates p_{xy} and is invisible at one cardinal station. A second station is justified when a known transport rotation makes that missing tilt or another contamination mode observable.

4 Particles following different paths

The single-rotation model assumes every particle samples the same magnetic fields and follows the same orbit. A real beam has transverse position, angle, momentum, and time coordinates ξ. Each trajectory may therefore have its own rotation $R(\xi)$. The state measured at station 2 is an ensemble average,

$$\overline{\boldsymbol{p}}_2 = \int d\xi\, f(\xi) R(\xi) \boldsymbol{p}_1, \tag{7}$$

$$\overline{\boldsymbol{P}}_2 = \int d\xi\, f(\xi) R(\xi) \boldsymbol{P}_1 R^{\mathsf{T}}(\xi). \tag{8}$$

The averaged vector and tensor maps are generally not rotations: they need not be orthogonal and can contract the polarization. Coherent rotation and phase spread are therefore different physical effects.

For illustration, if all trajectories rotate about the local beam z axis but their rotation angle has a Gaussian RMS spread σ_ψ, an azimuthal harmonic of order m is reduced by

$$\left\langle e^{im\psi} \right\rangle = e^{im\langle\psi\rangle} e^{-m^2\sigma_\psi^2/2}. \tag{9}$$

Vector first harmonics are suppressed by $e^{-\sigma_\psi^2/2}$, while tensor second harmonics are suppressed more rapidly, by $e^{-2\sigma_\psi^2}$. This can look like depolarization even though every individual particle undergoes a perfectly coherent spin rotation.

Trajectory variation also affects the detector response itself:

- beam offsets and angles shift the local detector azimuths and mix the $\sin\phi$, $\cos\phi$, $\sin 2\phi$, and $\cos 2\phi$ harmonics;
- momentum-dependent spin phase can correlate polarization with acceptance;
- apertures and losses can change the trajectory weights between stations;
- a trajectory-dependent efficiency can generate false L/R or U/D asymmetries.

The measured channel mean is therefore more accurately written as

$$\mu_k = \int d\xi\, f(\xi)\, w_k(\xi)\, \sigma_k \Big[R(\xi)\boldsymbol{p}_1,\, R(\xi)\boldsymbol{P}_1 R^{\mathsf{T}}(\xi) \Big], \tag{10}$$

where $w_k(\xi)$ contains orbit acceptance and detector efficiency.

If $f(\xi)$ and $R(\xi)$ are unknown, their parameters become nuisances and the effective polarization rank can decrease. Two polarimeters alone cannot recover an arbitrary path distribution. A defensible two-station analysis should use:

1. an orbit and spin transfer map over the measured beam phase space;
2. beam-position, angle, and momentum diagnostics at both stations;
3. trajectory-binned or event-by-event response calculations where possible;
4. finite detector acceptance and calibrated relative efficiencies;
5. ensemble-transport parameters included in the Fisher or likelihood fit.

5 Practical conclusion

One well-calibrated, multi-angle L/R/U/D station does not reconstruct a general spin–1 density matrix. Under favorable analyzing-power and normalization conditions it determines six of eight vector-plus-tensor degrees of freedom; p_z and p_{xy} remain exact nulls. For the present tensor-focused pure-p_{yy} program, this is sufficient for the primary after-SRC magnitude and stability measurement.

Two stations provide genuinely new information only when the known relative spin transport makes their null spaces complementary. A generic rotation with a transverse axis can, in principle, give full eight-dimensional density-matrix rank. Non-uniform particle paths replace that rotation by an ensemble map and can reduce polarization and identifiability. The scientific case for a second station should therefore be based on a phase-space-dependent spin-transfer model and a quantified reduction of the combined null space, not only on increased statistics.